Status	Finished
Started	Saturday, 17 October 2026, 11:17 AM
Completed	Saturday, 17 October 2026, 11:22 AM
Duration	4 mins 57 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Combining sorted sublists
c. Dividing the input into smaller subproblems
d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting
b. Sorting data in a single pass
c. Dividing the input into parts, solving each part, and combining the solutions
d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **3**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It makes the data unsorted
- b. It makes it easier to select items based on their relationship to the rest of the items
- c. It complicates the selection of items
- d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **4**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It recursively sorts subproblems
- b. It divides the list into sublists
- c. It uses the heap data structure
- d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **5**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. arrange()
- c. order()
- d. sorted()

The correct answer is: sort()

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Binary Search
- b. Linear Search
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It requires external libraries
- b. It sorts data out of the box efficiently
- c. It is less efficient than custom sorting algorithms
- d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Merge sort
- b. Heap sort
- c. Selection sort
- d. Quick sort

The correct answer is:
Merge sort

Question **9**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Merging
b. Rearranging
c. Sorting
d. Searching

The correct answer is: Sorting

Question **10**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are split again into smaller sublists
c. They are combined to form a single sorted list
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting helps in finding duplicates quickly
b. Sorting is rarely used in programming
c. Sorting decreases the efficiency of selection operations
d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **12**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Arranging data in a particular format
- b. Deleting data from a list
- c. Inserting data into a list
- d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **13**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is searched for a specific element
- b. Each part is sorted independently
- c. The parts are merged without sorting
- d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It divides the list into two halves, sorts each half, and then merges them
- b. It builds a sorted array one element at a time
- c. It compares adjacent elements and swaps them if necessary
- d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **15**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()
b. order()
c. sort()
d. sorted()

The correct answer is: sorted()